

# GMC representations and the intensity conjecture

A side note on construction versus selection

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**Position.** The Gaussian multiplicative chaos (GMC) representation does not, by itself, change the leading conjecture

$$\rho_\beta(z) = \frac{4/\beta - 1}{\pi(1 - |z|^2)^2}. \quad (1)$$

It provides an explicit family of hyperbolically invariant processes, including a candidate with this intensity, and exposes a precise selection problem. Constructing that candidate is not a proof that the finite disk gas converges to it. This note records those distinctions. It uses the random Bergman construction developed in the preceding discussion as an input; it does not reproduce its analytical proof.

## 1 The finite gas uses a biased Gaussian law

Let  $\mu = dA/\pi$  on the unit disk, and consider the labelled finite gas

$$d\mathbb{P}_{N,\beta}(\mathbf{z}) = Z_N(\beta)^{-1} |\Delta_N(\mathbf{z})|^\beta d\mu^N(\mathbf{z}), \quad \Delta_N = \prod_{i < j} (z_i - z_j).$$

Throughout this note,  $0 < \beta < 2$  and  $t = 2 - \beta \in (0, 2)$ . Take a centered real Gaussian generalized field  $X$  with covariance  $C(z, w) = -\log |z - w|$ , and let

$$M_t^X(dz) = :e^{\sqrt{t}X(z)}: d\mu(z).$$

The colons denote Wick normalization in the GMC limit. For a finite measure  $M$  with full support, define

$$G_N(M)_{jk} = \int z^j \bar{z}^k M(dz), \quad 0 \leq j, k < N, \quad D_N(M) = N! \det G_N(M).$$

The determinant identity and the GMC product-moment identity give

$$D_N(M) = \int |\Delta_N|^2 dM^{\otimes N}, \quad \mathbb{E} D_N(M_t^X) = Z_N(2 - t).$$

Hence the exact mixing law for the original gas is

$$\frac{dQ_{N,t}}{dP_{\text{GMC}}}(M) = \frac{D_N(M)}{\mathbb{E} D_N(M_t^X)}. \quad (2)$$

Conditionally on  $M$ , sample the rank- $N$  polynomial projection DPP in  $L^2(M)$ . Its labelled law is  $|\Delta_N|^2 dM^{\otimes N}/D_N(M)$ . Averaging with (2) cancels  $D_N$  and recovers  $\mathbb{P}_{N,2-t}$  exactly.

In contrast, averaging the same conditional DPP with the original Gaussian law  $P_{\text{GMC}}$  defines the *unbiased mixture*. Its asymptotics do not automatically describe the gas. Although  $D_N$  is a normalization factor inside the conditional particle law, it is a nonconstant random weight in the law of the environment.

## 2 What the infinite GMC construction supplies

Let  $G$  be the zero-boundary Gaussian free field with covariance

$$g_{\mathbb{D}}(z, w) = -\log \left| \frac{z - w}{1 - z\bar{w}} \right|, \quad M_t^G = :e^{\sqrt{t}G}: d\mu.$$

For  $s > -1$ , put

$$m_{t,s}(dz) = (1 - |z|^2)^s M_t^G(dz).$$

Write  $\mathcal{Y}_{t,s}$  for the following position law: first sample  $M_t^G$ , then sample the infinite projection DPP corresponding to the closure of  $\mathbb{C}[z]$  in  $L^2(m_{t,s})$ .

**Construction result used here.** For each fixed  $t \in (0, 2)$  and  $s > -1$ , this projection is locally trace class almost surely; its finite polynomial projections converge locally. The annealed process  $\mathcal{Y}_{t,s}$  is locally finite, almost surely infinite, and hyperbolically invariant, with ordinary-area intensity

$$\rho_{t,s}^{\mathcal{Y}}(z) = \frac{s+1}{\pi(1-|z|^2)^2}. \quad (3)$$

The coefficient is independent of  $t$  at fixed  $s$ ; the entire law is not asserted to be independent of  $t$ . The proof uses bounded analytic point evaluations from GMC small-ball estimates, followed by an equivariant polar decomposition and hyperbolic mass transport. Standard negative GMC moments are one input; see [1].

To relate this family to the unbiased mixture, decompose

$$X = G + \operatorname{Re} F, \quad \mathbb{E}F(z)\overline{F(w)} = -2\log(1 - z\bar{w}),$$

where  $G$  and the analytic Gaussian field  $F$  are independent. Wick normalization gives

$$M_t^X = |h|^2(1 - |z|^2)^{t/2} M_t^G, \quad h = e^{\sqrt{t}F/2}.$$

The harmonic factor can be removed from the infinite polynomial DPP. Precisely, with  $m_0 = (1 - |z|^2)^{t/2} M_t^G$ ,

$$\overline{h\mathbb{C}[z]}^{L^2(m_0)} = \overline{\mathbb{C}[z]}^{L^2(m_0)} \quad \text{almost surely.}$$

This density statement is essential because  $h$  need not be bounded. It follows by approximating  $h$  and 1 using  $h(rz)$  and  $h(z)/h(rz)$  in  $L^2(m_0)$  along a suitable sequence  $r \uparrow 1$ . Multiplication by  $h$  then gives a unitary identification, and its kernel factors cancel the change of reference measure in every DPP correlation measure. Thus the unbiased infinite mixture is  $\mathcal{Y}_{t,t/2}$ , with

$$\rho_{\beta}^{\text{unbiased}}(z) = \frac{1+t/2}{\pi(1-|z|^2)^2} = \frac{2-\beta/2}{\pi(1-|z|^2)^2}. \quad (4)$$

## 3 Construction does not verify the conjecture

Let  $b$  denote the coefficient multiplying  $1/[\pi(1-|z|^2)^2]$ . The relevant comparison is

| Position law               | Exponent $s$ | Coefficient $b$                  |
|----------------------------|--------------|----------------------------------|
| Unbiased GMC limit         | $t/2$        | $1 + t/2$                        |
| GMC candidate matching (1) | $2t/(2-t)$   | $(2+t)/(2-t)$                    |
| Actual disk-gas limit      | Unidentified | Unproved at fixed $\beta \neq 2$ |

The actual limit is not assumed to belong to the  $\mathcal{Y}_{t,s}$  family. The two explicit coefficients satisfy

$$\frac{b_{\text{conjectured}}}{b_{\text{unbiased}}} = \frac{2}{\beta}, \quad b_{\text{conjectured}} - b_{\text{unbiased}} = \frac{(2-\beta)(4-\beta)}{2\beta} > 0.$$

Therefore removing the determinant weight cannot establish the proposed coefficient. This comparison does *not* prove that the actual fixed-temperature gas limit differs from the unbiased mixture: its intensity is precisely what remains unknown.

More importantly, (3) realizes *every* positive coefficient by choosing  $s = b - 1$ . The choice

$$s_*(\beta) = \frac{4}{\beta} - 2 = \frac{2(2-\beta)}{\beta} \tag{5}$$

is obtained by solving  $s + 1 = 4/\beta - 1$ . It builds the proposed intensity into a candidate; it is not an independent derivation of that intensity from the finite gas. Existence of many invariant candidates also does not show nonuniqueness of the finite- $N$  limit, or nonuniqueness for one fixed DLR specification. A common specification has not been established for this family.

## 4 Perturbative evidence and its limitation

In terms of  $t = 2 - \beta$ , the leading conjecture predicts

$$b_{\text{conjectured}} = \frac{2+t}{2-t} = 1 + t + \frac{t^2}{2} + O(t^3), \quad b_{\text{unbiased}} = 1 + \frac{t}{2}.$$

The previously obtained first temperature response of the finite gas is

$$\lim_{N \rightarrow \infty} \partial_\beta \rho_{N,\beta}(z)|_{\beta=2} = -\frac{1}{\pi(1-|z|^2)^2},$$

locally uniformly in the disk. The earlier second-response calculation likewise matches the quadratic term of the first expansion. These are inputs from the preceding calculations, not new derivative results of this side note.

They support retaining (1) as the leading conjecture. However, derivatives were taken *before*  $N \rightarrow \infty$ . Without a justified exchange of limits or a uniform temperature remainder, those results do not exclude the unbiased coefficient as a possible fixed-temperature limit. Pointwise convergence of analytic functions on a real interval alone does not imply convergence of their derivatives.

## 5 A sharper conjecture to test

**Conjecture 1** (Tentative GMC selection conjecture). *For each  $0 < \beta < 2$ , the point process of the original  $N$ -particle disk gas converges locally in distribution to*

$$\mathcal{Y}_{2-\beta, 2(2-\beta)/\beta}.$$

This is stronger than the intensity conjecture: it specifies the entire limiting law. If true, it would give full-sequence convergence, hyperbolic invariance, and (1) for the limiting process. Its plausibility must be tested independently through higher correlations, Palm laws, and conditional resampling laws, including the probabilities of different particle counts. Matching an intensity is insufficient. In particular, we have not proved that this candidate satisfies the required limiting DLR specification.

One exact way to formulate the remaining selection mechanism is through orthogonal-polynomial norms. Let

$$h_N(M) = \min_{\substack{p \text{ monic} \\ \deg p = N}} \int |p|^2 dM.$$

Since  $D_N = N! \prod_{j=0}^{N-1} h_j$ , the biased environment laws obey

$$\frac{dQ_{N+1,t}}{dQ_{N,t}}(M) = \frac{h_N(M)}{\mathbb{E}_{Q_{N,t}} h_N(M)}. \quad (6)$$

Identifying the effect of these successive weights on local particle observables is a concrete unresolved task. Fixed-environment convergence of the conditional DPP does not justify replacing  $Q_{N,t}$  by the unbiased law.

The original leading conjecture is therefore retained, while Conjecture 1 is an additional, more tentative proposal. All constructions in this note concern  $0 < \beta < 2$ ; the positive real GMC representation used here gives no direct conclusion for  $\beta > 2$ .

## References

- [1] C. Garban, N. Holden, A. Sepúlveda and X. Sun, *Negative moments for Gaussian multiplicative chaos on fractal sets*, Electron. Commun. Probab. 23 (2018), no. 100. [arXiv:1805.00864](https://arxiv.org/abs/1805.00864).